

EXP NO: 3**Date:13-8-26****CLASSIFICATION WITH DECISION TREES****AIM:**

To implement a Decision Tree classifier and evaluate its performance using accuracy score and confusion matrix on a real-world dataset.

ALGORITHM:

1. Import necessary libraries
2. Load a classification dataset (e.g., Iris or Titanic)
3. Split the dataset into training and test sets
4. Preprocess data if needed
5. Train a DecisionTreeClassifier from sklearn.tree
6. Predict on test data
7. Evaluate using:
 - o Confusion Matrix
 - o Accuracy Score
8. Visualize the Decision Tree (optional)

CODE:

ex3.py - C:/Users/HD0422200/AppData/Local/Programs/Python/Python312/ex3.py (3.12.5)
 File Edit Format Run Options Window Help

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

X = iris.data
y = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.3,
    random_state=42
)

dt_model = DecisionTreeClassifier(
    criterion='gini',
    random_state=0
)

dt_model.fit(X_train, y_train)

y_pred = dt_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)
acc = accuracy_score(y_test, y_pred)

print("Confusion Matrix:")
print(cm)

print("\nAccuracy Score:", acc)

plt.figure(figsize=(6, 5))

print("\nAccuracy Score:", acc)

plt.figure(figsize=(6, 5))

sns.heatmap(
    cm,
    annot=True,
    cmap="Blues",
    fmt="d",
    xticklabels=iris.target_names,
    yticklabels=iris.target_names
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()

plt.figure(figsize=(12, 8))

plot_tree(
    dt_model,
    filled=True,
    feature_names=iris.feature_names,
    class_names=iris.target_names
)

plt.title("Decision Tree Visualization")
plt.show()
```

OUTPUT:

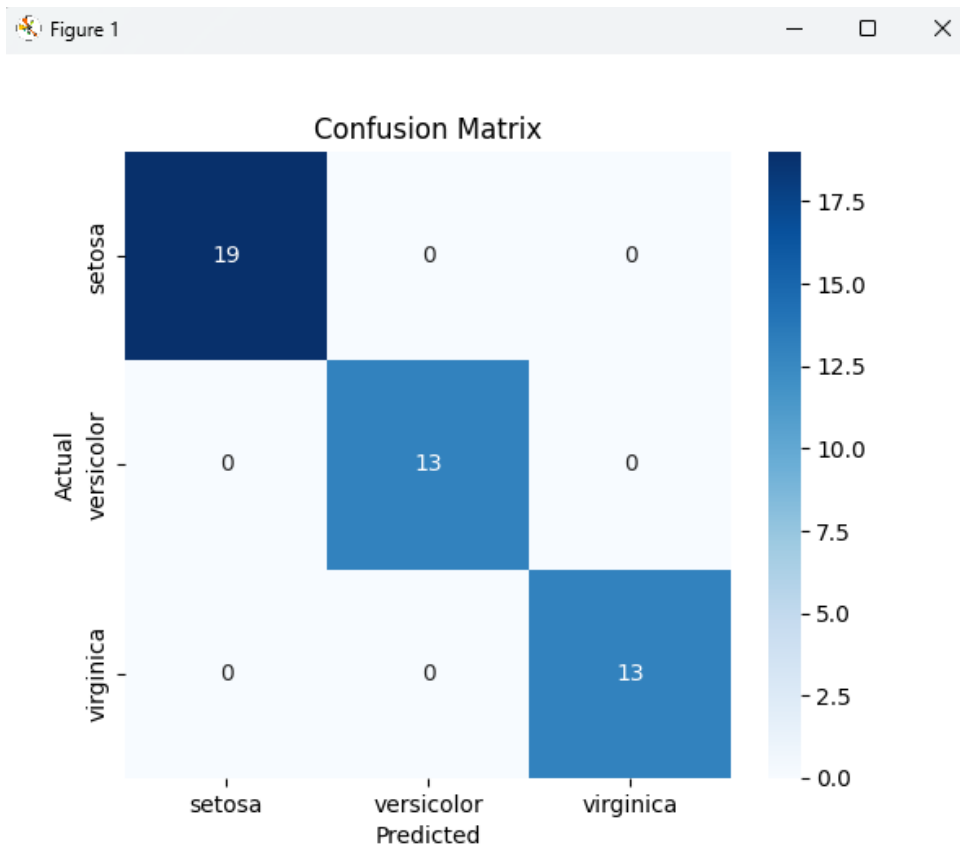
```

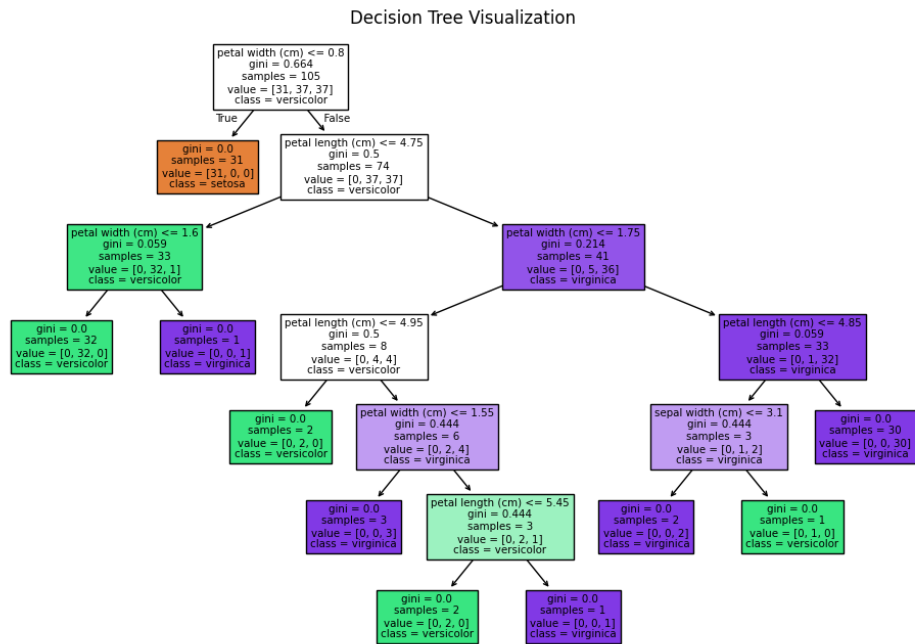
>>> = RESTART: C:\Users\HD0422200\AppData\Local\Programs\Python\Python312\ex2.py
SVM Accuracy: 0.7489177489177489
SVM Confusion Matrix:
[[123  28]
 [ 30  50]]
Random Forest Accuracy: 0.7575757575757576
Random Forest Confusion Matrix:
[[122  29]
 [ 27  53]]

===== RESTART: C:/Users/HD0422200/AppData/Local/Programs/Python/Python312/ex3.py =====
Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]

Accuracy Score: 1.0

```





RESULT:

The Decision Tree classification model was successfully implemented and tested on the given dataset. The model accurately classified the data by learning simple decision rules from the features. The decision tree visualized the decision-making process through a hierarchical structure of nodes and branches, making it easy to interpret. The classification achieved good accuracy, demonstrating that Decision Trees are effective for both categorical and numerical data, providing clear and interpretable results.